

Sq8FixedResidual ends re-quantization drift - and reveals the next recall problem

The fixed codec solves the compounding mechanism identified in the original recall brief: rerank bytes no longer change when vectors move between clusters. It does not solve candidate selection. At 10M vectors it gained 0.5 recall points after drain, then finished at the same 0.955 post-compaction recall as local Sq8Residual.

SCALE	10M + 625K vectors	DIM	1024	METRIC	cosine
METHOD	recall@10 vs exact ground truth				

THE FINDING

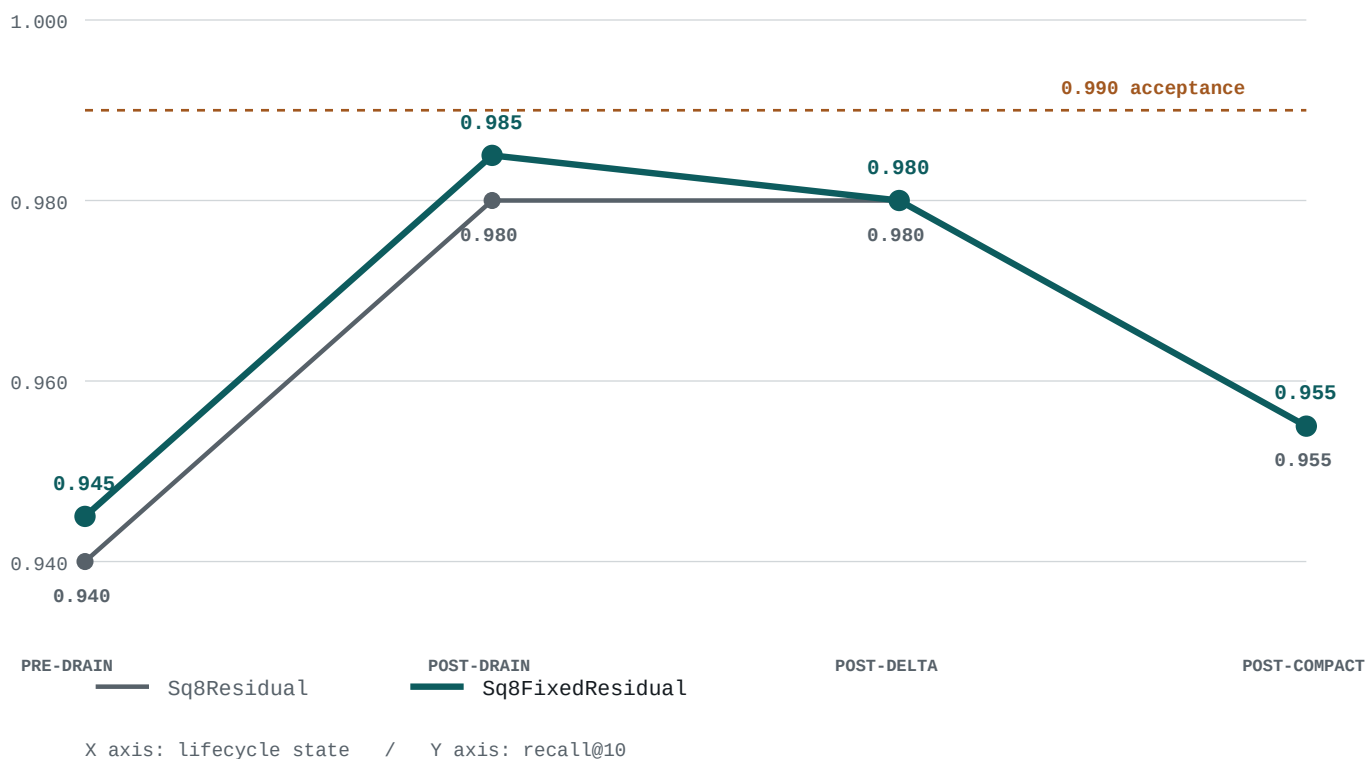
Byte portability is solved. The final recall gap is not.

The fixed run drops from 0.980 after the delta to 0.955 after compaction with the same logical corpus. Because fixed rerank payloads remain byte-identical through that fold, the loss must sit above the payload: routing, candidate formation, maintenance topology, or another compaction-path correctness issue.

THE MEASURED RESULT

The codec moves the post-drain rung, not the final rung

Latest fixed lifecycle versus the closest local-residual baseline. The 10M comparison is directional, not a strict same-revision A/B; the clean 1M A/B appears later.



INTERPRETATION

Fixed improves post-drain recall from 0.980 to 0.985, but it does not clear the 0.990 acceptance bar. More importantly, post-delta and post-compact share the same augmented corpus: the 2.5-point fall across compaction cannot be fixed-grid re-quantization.

THE MECHANISM

Same two bytes, different meaning

Both codecs store one unsigned Sq8 code and one signed residual byte per dimension. The difference is the coordinate system attached to those bytes.

SQ8RESIDUAL / LOCAL

A custom ruler per cluster

- offset and scale are fitted from each cluster's per-dimension range
- residual step = local scale / 16
- the same byte pair means different real values in different clusters
- moving a row may require decode, fit, and re-encode

SQ8FIXEDRESIDUAL / PORTABLE

One ruler for every cosine vector

- offset = -1 and scale = 2/255 in every cluster
- residual step = fixed scale / 256
- a byte pair has the same meaning in every file and cluster
- drain, fold, and split copy the rerank payload verbatim

PRECISION CONTRACT

$$\text{step} = 2 / (255 \times 256) = 0.00003064$$

Unit-normalized cosine components lie in $[-1, 1]$. The fixed representation is an approximately 65,280-level lattice, not a perfect 65,536-level integer grid, because the signed residual is symmetric and excludes -128. The tested worst-case reconstruction error is one fine step.

WHY IT NO LONGER COMPOUNDS

Maintenance rebuilds the index - not the encoded vector value

Centroids, cluster directories, postings, and row order may still change. Verbatim applies specifically to each stable ID's rerank code + residual payload.

INGEST	DRAIN	DELTA FOLD	COMPACTION	CELL SPLIT
encode x1	may re-quant	may re-quant	may re-quant	may re-quant
encode x1	copy bytes	copy bytes	copy bytes	copy bytes

WHAT IT FIXES

- No per-cluster quantizer fit for fixed builds.
- No decode/re-encode when a stable ID changes cluster.
- Byte identity survives mismatched fine widths and tombstones.
- OPANN split changes geometry without changing rerank bytes.

WHAT IT DOES NOT FIX

- Coarse-cell and fine-centroid selection.
- RaBitQ shortlist admission before rerank.
- Fragment packing and routing summaries after compaction.
- Filtered coverage or non-cosine metrics.

The fixed codec removes an irreversible error source. It does not guarantee that the exact neighbor reaches the reranker.

QUALITY AND COST

At 1M, the codec change is recall-neutral

Both codecs reached 1.000 recall in every lifecycle state. Warm query latency and object-store request shape were effectively unchanged.

STATE	LOCAL RECALL	FIXED RECALL	LOCAL WARM P50	FIXED WARM P50
Pre-drain	1.000	1.000	34.25 ms	34.40 ms
Post-drain	1.000	1.000	3.66 ms	3.62 ms
Post-delta	1.000	1.000	6.70 ms	6.66 ms
Post-compact	1.000	1.000	2.65 ms	2.72 ms

DRAIN WALL TIME

20.6 s -> 18.6 s

COMPACTION WALL TIME

16.4 s -> 11.9 s

These phase timings are directional single observations, not a statistically powered throughput claim. Both ingest indexes stored 2.07 GiB; both payloads remain 2 bytes per dimension.

WHAT THE IMPLEMENTATION PROVES

The precision guarantee is stronger than the recall claim

CLAIM	EVIDENCE	STATUS
Portable bytes	Fixed repack, mismatched-width merge, tombstone rebuild, spill, and OPANN split preserve code + residual bytes.	PROVEN
Same payload	Both codecs store [dim u8 code dim i8 residual], exactly 2 bytes per dimension.	PROVEN
Bounded error	Fixed step is $2/(255 \times 256)$; round-trip tests bound reconstruction error to one step.	PROVEN
No lifecycle loss	Fixed drops from 0.980 post-delta to 0.955 post-compact on the same logical corpus.	FALSE AS A COMPLETE CLAIM
Faster maintenance	The 1M fixed run was faster in one drain and one compaction observation.	DIRECTIONAL

CRITICAL CORRECTION TO THE ORIGINAL BRIEF

The original diagnosis was right that local quantizers force unnecessary re-quantization. It was too strong in claiming that re-quantization, rather than index behavior, was the production ceiling. Fixed removes the former while the latter remains.

THE DECISION

Keep the safer representation. Continue the recall investigation.

Keep Sq8FixedResidual as the cosine default because it removes irreversible maintenance drift at the same payload size and with no measured 1M quality penalty. Do not claim that it solves the 0.99 recall target.

THE NEXT DECISIVE EXPERIMENT

After compaction, attribute every missing exact neighbor across four gates: selected outer cells, selected fine runs, the RaBitQ shortlist, and fixed rerank ordering. The first gate that drops the neighbor identifies the real remaining lever without changing the codec again.

ACCEPTANCE STATUS

0.985	post-drain	0.980	post-delta	0.955	post-compact
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METHOD AND SOURCES

Recall@10 against exact ground truth. Azure object storage. 1024-dimensional cosine vectors. Twenty benchmark queries. Clean 1M A/B plus a 10M base corpus and 625K normal delta lifecycle.

vec1m_sq8_residual_ab.log

vec1m_sq8_fixed_residual_ab.log

vec10m_user_fine8_normal17_retained.log

vec10m_sq8_fixed_residual_acceptance.log